National University of Computer and Emerging Sciences

CS2005 - DATABASE SYSTEMS (S'26)

ASSIGNMENT # 3

Due Date: 16th April 2026

1. Failure to submit according to the above format would result in a deduction of 10% marks. Submissions other than Google Classroom (e.g., email, etc.) will not be accepted.
2. **Deadline:** The deadline for submitting the assignment is 10th April 2026**.** No late submission will be accepted. Correct and timely submission of the assignment is the responsibility of every student; hence no relaxation will be given to anyone.

3. **Honor Policy:** This assignment is a learning opportunity that will be evaluated based on your ability. Plagiarism cases will be dealt with strictly. If found plagiarized, both the involved parties will be awarded zero marks in this assignment, all the remaining assignments, or even an F grade in the course.

Part A: Relational Algebra Queries [40 Marks]

Assume you have normalized the "Galactic Operations" data into 1NF to 3NF. Use your derived schema (tables, keys, attributes) and write each query in formal relational algebra. Clearly state any assumptions (e.g. whether Pool_Size is numeric by removing "M"). For each query: give the RA expression and briefly explain your approach.

1. Join & Selection (4 pts): Find the *Operation_Code* and Director for all operations that use a *Tech_Asset* manufactured by 'ArmaTech' *and* are funded by 'FedFund' and have at least one *Personnel_Type* = 'Pilot'.
2. Join & Conditions (4 pts): List the *Personnel_ID* of all *Engineers* who have a Base_Salary > 100000 and who worked on any operation in a sector with *Threat_Level* = 'High'.
3. Division (6 pts): Find the *Personnel_ID* of personnel who participated in operations in every sector where *Threat_Level* = 'High'.
4. Division (6 pts): Find the *Funding_Source* of funding bodies that funded operations in every operation type (*Supply, Recon, Rescue, Patrol, Escort, Survey*).
5. Set Difference (5 pts): List the *Tech_Asset* names that were used in no operation funded by 'StellarGrant'.
6. Aggregation Logic (5 pts): Find the *Operation_Code* of any operation where the number of Engineers involved exceeds 2.
7. Self-Join (5 pts): Find *Operation_Code* of operations that involve two personnel with the same Last Name (hint: assume personnel IDs encode names, or use a fictitious LastName attribute). This checks self-join on Personnel. (If names are not explicit, treat *Personnel_ID* sharing prefix as same family, e.g. P101 and P115 both "Smith".)
8. Nested Join (5 pts): Retrieve the *Operation_Name* and *Assignment_Date* for operations where the funding's *Pool_Size* (remove "M" for numeric) is less than the smallest *Base_Salary* of any Technician on that operation. (Requires joining Operation, Funding, OperationPersonnel, Personnel and comparing numeric values.)

(Any valid RA equivalent is acceptable. Show reasoning for division-based queries.)

National University of Computer and Emerging Sciences

Part B: Data Normalization & Dependencies [60 Marks]

You are given the flat file (CSV) above. By analyzing it, you must deduce functional dependencies (FDs), clean anomalies, and normalize to 3NF. Show all steps.

1. **Functional Dependencies (10 pts):** List *all non-trivial FDs* you can infer from the dataset, then compute a minimal cover. For each FD in your minimal cover, cite at least three example rows that *witness* that FD (e.g. show the same left-hand values with a unique right-hand). *Hint:* Consider FDs from Operation_Code, Director, Sector, Tech_Asset, Vendor, Personnel_ID, and Funding_Source as determinants. (A minimal cover has single-attribute RHS and no redundancies.)
2. **Data Anomalies (5 pts):** Identify at least three inconsistencies or anomalies in the data. For each, state what the problem is (e.g. "Director 'Elara Voss' has different Clearance values"), and *propose a correction* or how you would model the data so it doesn't happen (e.g. add a Director table with unique Clearance). Explain how these anomalies violate 1NF/2NF rules.
3. **Normalization to 3NF (20 pts):** Decompose the flat table into relations in 1NF, 2NF, and finally 3NF. List each resulting table with its attributes, primary key, and foreign keys. Show which FD caused each decomposition (partial or transitive). Ensure *lossless-join* (demonstrate via a join diagram or argument) and that all original FDs are preserved. *Hint:* After 3NF, each non-key attribute should depend on the whole key or a key. (You may assume 1NF by treating each row as atomic.)
4. **Conditional Dependencies (10 pts):** Some dependencies hold only under certain conditions or by business rule (they are not *unconditional* FDs). Identify at least two such dependencies implicit in the data (for example, "Engineers cannot have Asset_Type = Weapon" or "If Personnel_Type = Medic then Threat_Level ∈ {Low, Medium}"). Explain how each is conditional (e.g. depends on context) and how it affects your design (should it be enforced as a constraint?).
5. **Proof of Correctness (15 pts):** For your 3NF design, prove (a) lossless-join: show that recomposing the tables yields the original (e.g. by join of keys); (b) dependency preservation: show that each FD from part (1) can be checked using just your decomposed tables. Finally, give one concrete example of an *update anomaly* (insertion, deletion, or update) that existed in the original flat file and show how your normalized design eliminates it.